Examining the Influence of Progenitor Parameters on Early-Time Emissions in the u, uv, and g Bands of Type IIP Supernovae

Arvind Chandrasekar

***Abstract*—We model the early-time emission in Type IIP supernovae to determine the variation of luminosity and flux in CASTOR's target *uv*, *u*, and *g* bands with progenitor properties such as radius (10^{12}-10^{14} cm), explosion energy (10^{50}-10^{52} erg), and mass (5-20 $M_{\odot}$). Using an analytic relation, we compute each supernova's bolometric luminosity and use it to derive a blackbody temperature. We employ the Planck function to determine the spectral radiance across each of our bands for various progenitors. The results suggest a higher detectability of larger progenitors, with larger radii and higher explosion energies allowing for greater luminosity in CASTOR's bands, consistent with Wien's displacement law. A greater mass is seen to lead to lower luminosities and fluxes in each band, despite influencing a spectral shift towards larger wavelengths.**

I. Introduction

Type IIP supernovae are a distinct class of core-collapse supernovae, identified by a characteristic plateau phase in their light curves after shock breakout [1]. At shock breakout, the supernova's shock wave travels through and reaches the boundary of its stellar surface, causing a short-lived burst of radiation, typically in the soft X-ray or UV region [2]. In this procedure, we determine the early detectability of Type IIP supernovae by the Canadian Space Agency's CASTOR (Cosmological Advanced Survey Telescope for Optical and ultraviolet Research) telescope [4] in its target bands - particularly at shock breakout. This is achieved by gauging the variation in absolute luminosities, using typical values of progenitor radii, masses, and explosion energies in the uv (150 - 300 nm), u (300 - 400 nm), and g (400 - 500 nm) bands. We first use a simplified analytic luminosity relation [3] to calculate the bolometric luminosity of a number of typical progenitors, with varying parameters (radius, explosion energy, and mass). We then assume a blackbody model, and use the Stefan-Boltzmann to compute each of their surface temperatures. We employ the Planck function [5] and integrate across our chosen uv, u, and g bands. This allows us to observe the temperature-dependent spectral flux density in each band, as driven by our chosen progenitor properties. An analysis into the impact on band-specific luminosity of varying multiple progenitor properties is also presented.

II. Theory and Methods

The bolometric luminosity of a Type IIP supernova at the instant of shock breakout (at $t = 0$) can be determined through the following (normalised) equation, as described by Arnett (1979)[3]:

$$L(1,0) = K_0 \left[\frac{R(0)}{10^{14}\ \text{cm}}\right] \left[\frac{E_{\text{Th}}(0)}{10^{51}\ \text{ergs}}\right] \left(\frac{0.4\ \text{cm}^2\ \text{g}^{-1}}{\kappa}\right) \left(\frac{1 M_{\odot}}{M}\right) \tag{1}$$

where L(1,0) describes the luminosity $L(x,t)$ where $x = 1$ is the dimensionless radial constant representing the photospheric surface, and $t = 0$ refers to the instant of implosion. K_o is a luminosity normalisation factor, taken to be 5.2×10^{43} (at $A = 0$, the uniform density case), with $R(0)$ being the progenitor radius, $E_{\text{Th}}(0)$ being the (thermal) explosion energy, κ as the opacity, and $M_{\odot}$ as the progenitor mass - with all of them normalised to certain respective typical values.

Our controlled variables of progenitor radius, mass, and explosion energies are each varied (with the other two parameters constant), for which we use the above equation to compute discrete values for the bolometric luminosity for different progenitors. Our parameter ranges can be seen in Table 1 below:

TABLE I
Ranges of Progenitor Parameters

Parameter	Range
Mass ($M_{\odot}$)	5 – 20
Radius (cm)	10^{12} – 10^{14}
Thermal Explosion Energy (ergs)	10^{50} – 10^{52}

With these, we use the rearranged Stefan-Boltzmann law below to derive the temperature of the photosphere at shock breakout for each progenitor, based on its luminosity and radius:

$$T = \left(\frac{L}{4\pi R^2 \sigma}\right)^{\frac{1}{4}} \tag{2}$$

Given the temperatures, we use the Planck function as a function of wavelength below to find spectral radiance in each of CASTOR's uv, u, and g bands:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}. \tag{3}$$

To find the flux in each band, this radiance is first converted to flux density and is then scaled by the emitting surface area. The expression is integrated over the limits of the chosen band to determine the band-specific luminosity.

$$L_{\text{band}} = 4\pi R^2 \int_{\lambda_{\min}}^{\lambda_{\max}} \pi B(\lambda, T)\, d\lambda. \tag{4}$$

Concurrently, we also use the Wein's displacement law as a continuous sanity check for the expected peak in each progenitor's emission, which we compare to the behaviour in each band in our plots.

III. Results and Discussion

A. Variations in Radius

Fixing the values of M at $10M\odot$ and E_{Th} at 10^{51} ergs, we vary R as 10^{12} cm, 10^{13} cm (our base case), and 10^{14} cm. The results are seen in Figures 1, 2, and 3 and in Table 2.

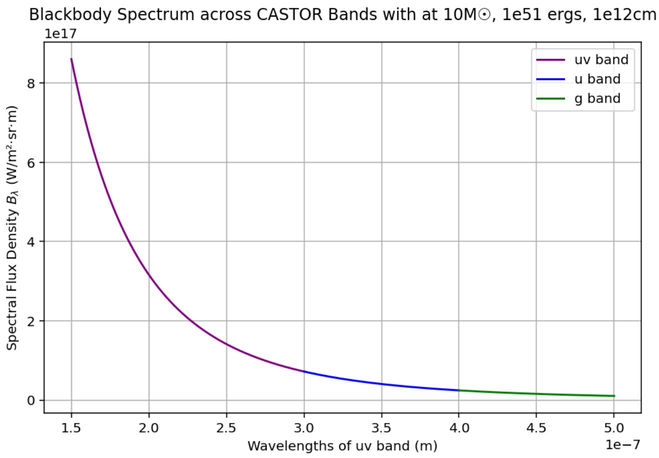

Fig. 1. Blackbody Spectrum across CASTOR Bands at 10^{12} cm

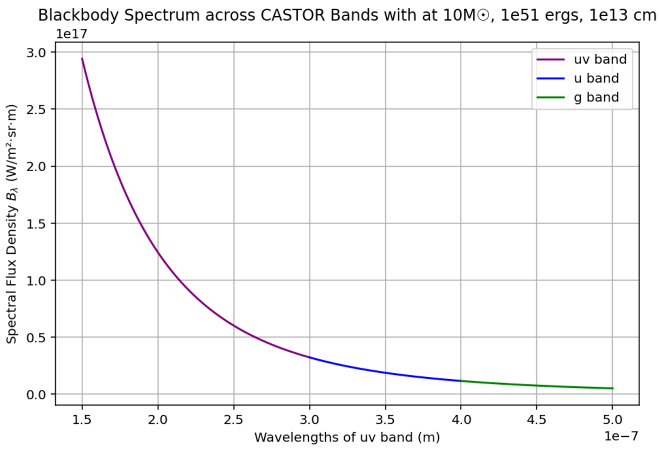

Fig. 2. Blackbody Spectrum across CASTOR Bands at 10^{13} cm

TABLE II
Radiative properties for models with varying progenitor radius at fixed mass 10 $M_\odot$ and explosion energy 10^{51} ergs

Radius	10^{12} cm	10^{13} cm	10^{14} cm
Temperature (K)	92,400	52,000	29,200
$L_{\rm bol}$ (W)	5.20×10^{33}	5.20×10^{34}	5.20×10^{35}
$L_{\rm uv}$ (W)	1.68×10^{32}	6.46×10^{33}	1.75×10^{35}
$L_{\rm u}$ (W)	1.70×10^{31}	7.84×10^{32}	3.03×10^{34}
$L_{\rm g}$ (W)	6.41×10^{30}	3.11×10^{32}	1.32×10^{34}
Peak wavelength (nm)	31.4	55.8	99.2

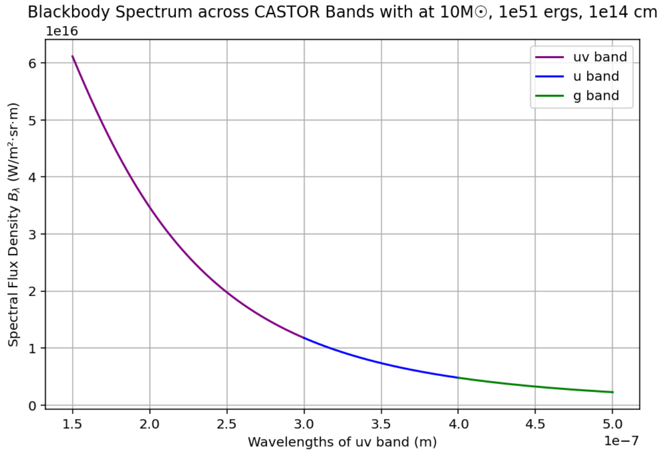

Fig. 3. Blackbody Spectrum across CASTOR Bands at 10^{14} cm

In accordance with blackbody theory and Equation 1, we note a decrease in temperature and an increased bolometric luminosity with greater progenitor radii. This is consistent with the $L \propto R^2T^4$ scaling, where an increase in surface area compensates for a decrease in temperature. Moreover, the peak wavelength of the emission at shock breakout increases with radius (consistent with Wien's displacement law) and shifts closer to our observed uv band; with a notably higher fraction of the bolometric luminosity being the uv band itself - from 3.22% of the bolometric luminosity at 10^{12} cm to 33.7% at 10^{14} cm. The general peak of emission in the range of XUV and UV-C bands (33 - 100 nm) points to our closest uv band (150 - 300 nm) being the most fruitful band for detectability. We further observe that the u and g bands also occupy a greater bolometric fraction with an increase in radius, underscoring the idea that a larger progenitor exhibits greater luminosity in CASTOR's target bands - with a particular increase in u and g bands - and thus greater detectability at shock breakout.

B. Variations in Thermal Explosion Energies

With constant values of M at $10M\odot$ and R at 10^{13} cm, we vary E_{Th} as 10^{50} ergs, 10^{51} ergs (the base case), and 10^{52} ergs. The base case is reflected in Figure 2, and the other two variations of E_{Th} can be seen in Figures 4 and 5, with the final results in Table 3 below.

TABLE III
Radiative properties for models with varying explosion energy at fixed radius 10^{13} cm and mass 10 $M_\odot$

Explosion Energy	10^{50} erg	10^{51} erg	10^{52} erg
Temperature (K)	29,200	52,000	92,400
$L_{\rm bol}$ (W)	5.20×10^{33}	5.20×10^{34}	5.20×10^{35}
$L_{\rm uv}$ (W)	1.75×10^{33}	6.46×10^{33}	1.68×10^{34}
$L_{\rm u}$ (W)	3.03×10^{32}	7.84×10^{32}	1.70×10^{33}
$L_{\rm g}$ (W)	1.32×10^{32}	3.11×10^{32}	6.41×10^{32}
Peak wavelength (nm)	99.2	55.8	31.4

As seen in Table 3, we notice an identical inversion in the bolometric and band-specific luminosities with an increase in explosion energy of the progenitor as opposed to an increase in radius. As expected, a higher (thermal) explosion energy is

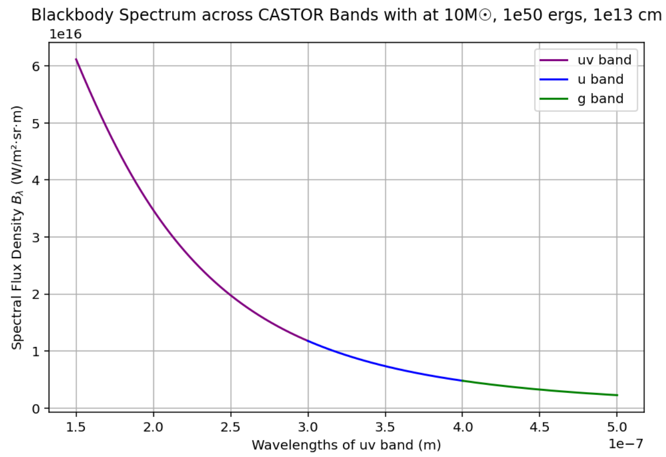

Fig. 4. Blackbody Spectrum across CASTOR Bands at $10^{50} E_{Th}$

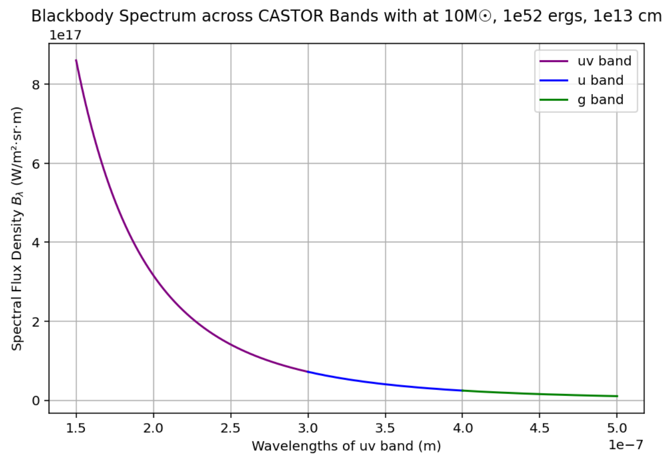

Fig. 5. Blackbody Spectrum across CASTOR Bands at $10^{52} E_{Th}$

synonymous with a higher progenitor surface temperature at shock breakout - which gives rise to lower peak wavelengths and a general shift into lower wavelength ranges (as low as 31.3 nm), away from CASTOR's target bands and causing lower luminosity fractions in each band. This is despite the general increase in the absolute luminosity in u, uv, and g, as brought about by the increase in energy. The bolometric luminosity scales linearly with energy (with an order of magnitude increase for the same change in the explosion energy) and this increase in absolute luminosity for all our target bands suggests an enhanced detectability as a result of higher values of E_{Th}.

C. Variations in Mass

The final parameter we vary is the progenitor mass M as $5M\odot$, $10M\odot$ (the base case), and $20M\odot$, with R at 10^{13} cm, and E_{Th} at 10^{51} ergs. Once again, the base case is shown in Figure 2 from earlier, and the other two variations of M can be seen in Figures 6 and 7, with the final results in Table 4.

In Table 4, we observe that the bolometric luminosity is a fourth of its original value at $5M\odot$ when the mass is quadrupled to $20M\odot$. This is likely due to the opacity imposed by an optically thicker envelope resulting from heavier progenitors [1]. Similarly, the temperature decreases by about 70% for the same change in mass, shifting the peak wavelength closer to our target bands - although it still doesn't peak in any of them. We ultimately infer that lighter progenitors are more favourable candidates for observation by CASTOR during shock breakout given their tendency to produce larger absolute values in our target uv, u, and g bands - despite a greater mass being the cause for a higher percentage of the bolometric luminosity being occupied by each band.

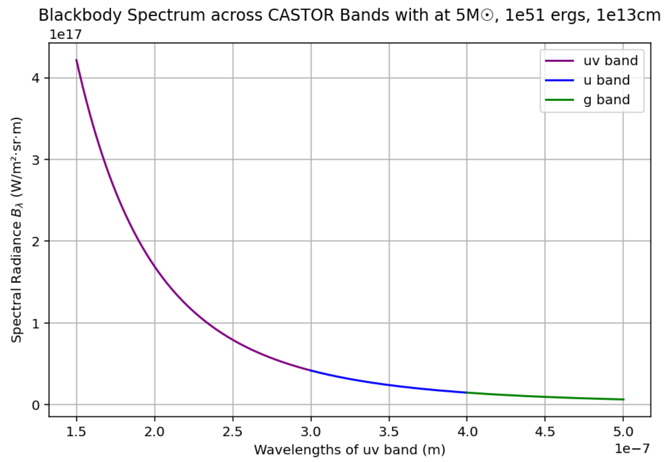

Fig. 6. Blackbody Spectrum across CASTOR Bands at $5M\odot$

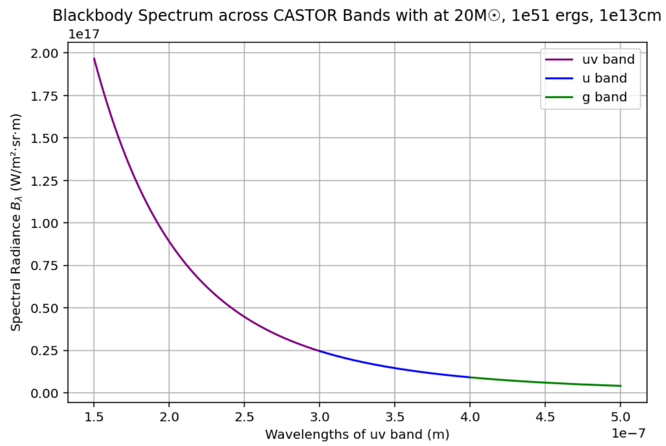

Fig. 7. Blackbody Spectrum across CASTOR Bands at $20M\odot$

TABLE IV
RADIATIVE PROPERTIES FOR MODELS WITH VARYING MASS AT FIXED RADIUS 10^{13} CM AND EXPLOSION ENERGY 10^{51} ERGS

Mass	$5\,M_\odot$	$10\,M_\odot$	$20\,M_\odot$
Temperature (K)	61,800	52,000	43,700
L_{bol} (W)	1.04×10^{35}	5.20×10^{34}	2.60×10^{34}
L_{uv} (W)	8.84×10^{33}	6.46×10^{33}	4.57×10^{33}
L_{u} (W)	1.00×10^{33}	7.84×10^{32}	6.03×10^{32}
L_{g} (W)	3.90×10^{32}	3.11×10^{32}	2.44×10^{32}
Peak wavelength (nm)	46.9	55.8	66.3

D. Luminosity in Target Bands as a Function of Multiple Parameters

We explore and visualise the impact on band luminosity of varying two of our three chosen progenitor properties at once through the following contour plots, which inform us of optimal parameter combinations for detectability at shock breakout. The luminosity in each of Figures 8, 9, and 10 is taken to be the normalised sum (i.e. normalised with respect to its maximum possible value) of the luminosity in the uv, u, and g bands. The radius and energy axes are plotted as log scales to improve the interpretability of the plots.
The results in Fig. 8 are evidently consistent with earlier plots and data, and reflect the greatest total luminosity (in CASTOR's target bands) and highest detectability for high thermal explosion energies and large progenitor radii.

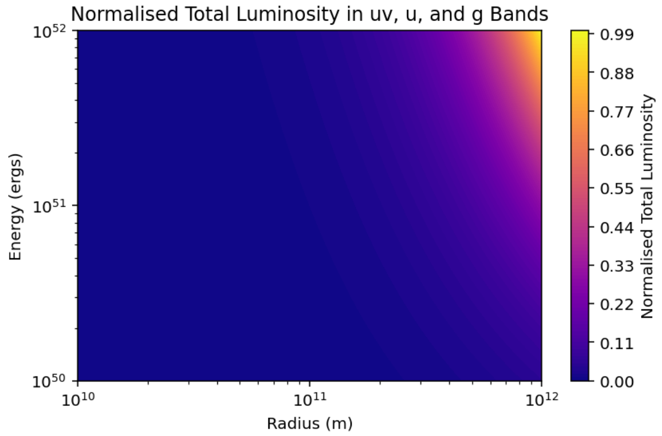

Fig. 8. Normalised Total Luminosity in Target Bands Maximised at High E_{Th} and High R

Fig. 9 below also corresponds with earlier predictions, indicating the greatest luminosity and detectability for larger and less massive progenitors, owing to the aforementioned influence of surface area scaling and mass-influenced photospheric opacities.

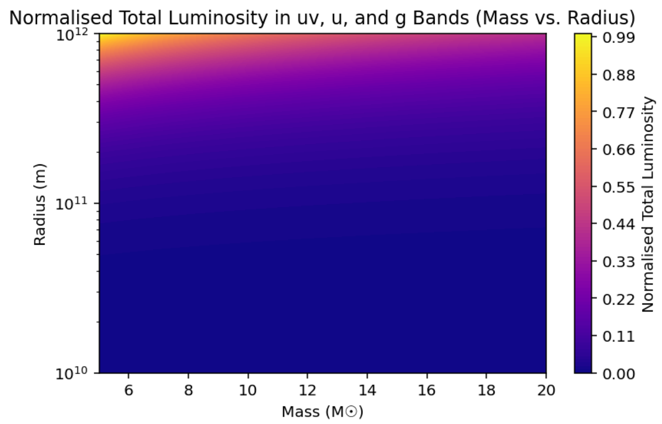

Fig. 9. Normalised Total Luminosity in Target Bands Maximised at High R and Low M

In Fig. 10, the luminosity and hence the detectability are maximised for high-energy, low-mass progenitors - in agreement with our computed plots and our understanding of the significance of mass and energy in shock breakout emissions.

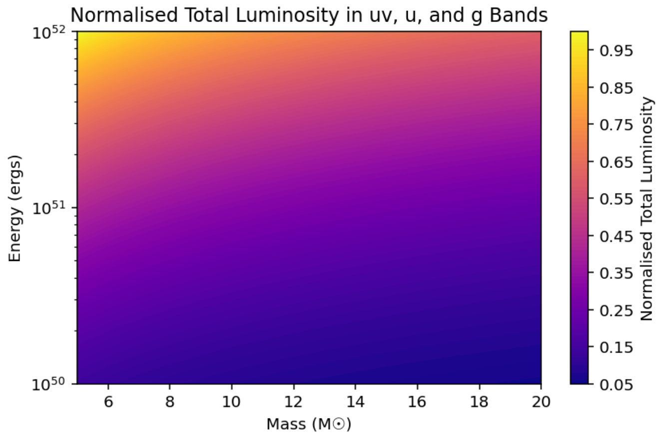

Fig. 10. Normalised Total Luminosity in Target Bands Maximised at High E_{Th} and Low M

Excellent agreement between the contour plots and computed results thus reinforces the validity of our analysis.

E. Assumptions and Constraints

It is important to note our assumption of a constant photospheric opacity of 0.4 cm²/g and ideal blackbody emission, which may not hold without local thermal equilibrium or line-dominated atmospheres. Moreover, we assume ideal bandpass efficiencies, which is unlikely to be the case in real observations.

IV. CONCLUSION

We note the immense contributions of progenitor properties of radius, mass, and explosion energies on the luminosity and flux of Type IIP supernovae, particularly in the u, uv, and g bands. We first observe the increased detectability of supernovae - as a result of a shift in wavelengths towards the observed bands - with larger progenitor radii. An increase in thermal explosion energy also generally warrants a greater luminosity in each of CASTOR's bands despite their occupation of a lower fraction of the bolometric luminosity (resulting from the wavelength shift towards lower frequencies corresponding to XUV bands). On the other hand, progenitor masses are seen to exhibit an inverse trend, with lower masses enhancing luminosity and hence detectability across all bands in general. In general, our results parametrise the highest detectability of progenitors - with large radii, low masses, and high thermal explosion energies being the conditions that enhance early-time transient observability in CASTOR's target bands.

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